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compact1, compact2, compact3

java.io

Class OutputStreamWriter

java.lang.Object
    java.io.Writer
        java.io.OutputStreamWriter

All Implemented Interfaces:

`Closeable`, `Flushable`, `Appendable`, `AutoCloseable`

Direct Known Subclasses:

`FileWriter`

```
public class OutputStreamWriter
extends Writer
```

An OutputStreamWriter is a bridge from character streams to byte streams: Characters written to it are encoded into bytes using a specified `charset`. The charset that it uses may be specified by name or may be given explicitly, or the platform's default charset may be accepted.

Each invocation of a write() method causes the encoding converter to be invoked on the given character(s). The resulting bytes are accumulated in a buffer before being written to the underlying output stream. The size of this buffer may be specified, but by default it is large enough for most purposes. Note that the characters passed to the write() methods are not buffered.

For top efficiency, consider wrapping an OutputStreamWriter within a BufferedWriter so as to avoid frequent converter invocations. For example:

```
Writer out
  = new BufferedWriter(new OutputStreamWriter(System.out));
```

A *surrogate pair* is a character represented by a sequence of two `char` values: A *high* surrogate in the range '\uD800' to '\uDBFF' followed by a *low* surrogate in the range '\uDC00' to '\uDFFF'.

A *malformed surrogate element* is a high surrogate that is not followed by a low surrogate or a low surrogate that is not preceded by a high surrogate.

This class always replaces malformed surrogate elements and unmappable character sequences with the charset's default *substitution sequence*. The CharsetEncoder class should be used when more control over the encoding process is required.

Since:

JDK1.1

See Also:

BufferedWriter, OutputStream, Charset

Field Summary

Fields inherited from class java.io.Writer

lock

Constructor Summary

Constructors

Constructor	Description
`OutputStreamWriter(OutputStream out)`	Creates an OutputStreamWriter that uses the default character encoding.
`OutputStreamWriter(OutputStream out, Charset cs)`	Creates an OutputStreamWriter that uses the given charset.
`OutputStreamWriter(OutputStream out, CharsetEncoder enc)`	Creates an OutputStreamWriter that uses the given charset encoder.
`OutputStreamWriter(OutputStream out, String charsetName)`	Creates an OutputStreamWriter

	that uses the named charset.

Method Summary

Concrete Methods

Modifier and Type	Method	Description
void	**close**()	Closes the stream, flushing it first.
void	**flush**()	Flushes the stream.
String	**getEncoding**()	Returns the name of the character encoding being used by this stream.
void	**write**(char[] cbuf, int off, int len)	Writes a portion of an array of characters.
void	**write**(int c)	Writes a single character.
void	**write**(**String** str, int off, int len)	Writes a portion of a string.

Methods inherited from class java.io.Writer

append, append, append, write, write

Methods inherited from class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructor Detail

OutputStreamWriter

```
public OutputStreamWriter(OutputStream out,
                          String charsetName)
                   throws UnsupportedEncodingException
```

Creates an OutputStreamWriter that uses the named charset.

Parameters:

out - An OutputStream

charsetName - The name of a supported charset

Throws:

UnsupportedEncodingException - If the named encoding is not supported

OutputStreamWriter

```
public OutputStreamWriter(OutputStream out)
```

Creates an OutputStreamWriter that uses the default character encoding.

Parameters:

out - An OutputStream

OutputStreamWriter

```
public OutputStreamWriter(OutputStream out,
                          Charset cs)
```

Creates an OutputStreamWriter that uses the given charset.

Parameters:

out - An OutputStream

cs - A charset

Since:

1.4

OutputStreamWriter

```
public OutputStreamWriter(OutputStream out,
                          CharsetEncoder enc)
```

Creates an OutputStreamWriter that uses the given charset encoder.

Parameters:

out - An OutputStream

enc - A charset encoder

Since:

1.4

Method Detail

getEncoding

```
public String getEncoding()
```

Returns the name of the character encoding being used by this stream.

If the encoding has an historical name then that name is returned; otherwise the encoding's canonical name is returned.

If this instance was created with the `OutputStreamWriter(OutputStream, String)` constructor then the returned name, being unique for the encoding, may differ from the name passed to the constructor. This method may return `null` if the stream has been closed.

Returns:

`The historical name of this encoding, or possibly null if the stream has been closed`

See Also:

`Charset`

write

```
public void write(int c)
           throws IOException
```

Writes a single character.

Overrides:

`write in class Writer`

Parameters:

`c - int specifying a character to be written`

Throws:

`IOException - If an I/O error occurs`

write

```
public void write(char[] cbuf,
                  int off,
                  int len)
           throws IOException
```

Writes a portion of an array of characters.

Specified by:

write in class Writer

Parameters:

cbuf - Buffer of characters

off - Offset from which to start writing characters

len - Number of characters to write

Throws:

IOException - If an I/O error occurs

write

```
public void write(String str,
                  int off,
                  int len)
           throws IOException
```

Writes a portion of a string.

Overrides:

write in class Writer

Parameters:

str - A String

off - Offset from which to start writing characters

len - Number of characters to write

Throws:

IOException - If an I/O error occurs

flush

```
public void flush()
           throws IOException
```

Flushes the stream.

Specified by:

flush in interface Flushable

Specified by:

flush in class Writer

Throws:

IOException - If an I/O error occurs

close

```
public void close()
           throws IOException
```

Description copied from class: `Writer`

Closes the stream, flushing it first. Once the stream has been closed, further write() or flush() invocations will cause an IOException to be thrown. Closing a previously closed stream has no effect.

Specified by:

`close` in interface `Closeable`

Specified by:

`close` in interface `AutoCloseable`

Specified by:

`close` in class `Writer`

Throws:

`IOException` - If an I/O error occurs

Submit a bug or feature
For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.
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